

The Demotion of Gauge Field Theory, the Theoretical Reconstruction of the Standard Model, and the Five-Layer Axiomatization Structure

— Precursor Cascade Asymmetry, the Regime Function of the Higgs Mechanism, the Statistical Correspondence of Virtual States, and the Final Formalization of the Factor Hierarchy Law (IV)

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Abstract

This paper is the concluding piece of the regime audit series on the Standard Model of particle physics, completing three core missions. First, it completes the demotion of gauge field theory — demonstrating that gauge field theory is not the most fundamental framework but rather the expression of regime boundaries in the language of group theory. The Factor Hierarchy Law is a deeper organizing principle than gauge field theory. Second, using regime boundary theory, it completes the theoretical reconstruction of the Standard Model. The statistical origin of CP violation lies in precursor cascade asymmetry — the cascade asymmetry between the strange→charm deviation of 0.0052 GeV and the charm→bottom deviation of 0.0000 GeV constitutes the statistical definition of symmetry breaking within the Chow test framework. Cascade asymmetry may be reactivated at cascade pairs that cross switching-type boundaries, with its complete verification requiring measurement of the bottom→top cascade deviation. The essential function of the Higgs mechanism is to generate regime boundaries — it partitions a continuous mass spectrum into discrete regime centers. Virtual intermediate states are the quantum expression of precursor signals. Third, based on all empirical findings from the previous three papers, it extracts directly usable mathematical assets — the charm-region piecewise decay equation, the bottom-region piecewise decay equation, the formalized expression of the precursor cascade, the universal formula for transition zone widths, and the intercept/slope ratio criterion for switching types. Fourth, it formalizes the entire theoretical system into a five-layer axiomatization structure — law layer, norm layer, algorithm layer, theorem-and-proposition layer, and theory layer — and proposes five testable predictions. Fifth, it completes the full audit of the Factor Hierarchy Law in the Standard Model of particle physics — a complete closed loop from empirical discovery to mathematical theorem to theoretical derivation to system architecture.

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1. Introduction

1.1 Complete Review of the Previous Four Papers

The first three papers of the regime audit series on the Standard Model of particle physics have already completed the full progression from empirical discovery to theoretical derivation.

Tang (2026n) used the Factor Hierarchy Law and Testability Norms to perform a zero-presupposition regime audit on 275 particles from the Particle Data Group (PDG) 2025 public dataset. An exhaustive scan of 11 activation operators automatically stratified into a charm region (~ 2.29 GeV) and a bottom region ($\sim 9.46\text{--}10.02$ GeV). The $W \times M^2$ activation operator amplified the Chow F statistic by 2569 times. The algorithm locked the positions of Λc^+ , $Y(1S)$, and $Y(2S)$ with 10^{-4} GeV precision.

Tang (2026o) further discovered the strange region ($\sim 1.4\text{--}1.6$ GeV) as the third regime center. The precursor cascade was verified with precision quantification — Strange $T_2 \approx$ Charm T_1 (deviation 0.0052 GeV), Charm $T_2 =$ Bottom precursor (deviation 0.0000 GeV). Three mathematical theorems — the Transition Zone Width Theorem (Theorem 1), the Switching Type Dichotomy Theorem (Theorem 2), and the Switching Type Intrinsic Property Theorem (Theorem 3) — were formally proved.

Tang (2026p) completed the theoretical derivation of the gauge group structure of the Standard Model from the Factor Hierarchy Law. The Gauge Group Structure Theorem (Theorem 4) proved that $SU(3)$ is the unique gauge group satisfying the three-state structure condition of the meta-regime factor. The Direct Product Theorem (Theorem 5) proved that the orthogonal independence of color charge and flavor quantum numbers necessarily leads to a direct product of gauge groups. The Three-Generation Proposition (Proposition 6) argued that the topological constraint of the transition zone width and the experimental exclusion facts from the LHC jointly point to the number of heavy quarks being limited to four.

1.2 The Threefold Mission of This Paper

The mission of this paper is threefold. First, to complete the demotion of gauge field theory — to demonstrate that gauge field theory is the expression of regime boundaries in the language of group theory, and that the Factor Hierarchy Law is a deeper organizing principle. Second, to use regime boundary theory

to complete a theoretical reconstruction of the Standard Model — reinterpreting CP violation, the Higgs mechanism, and virtual intermediate states. Third, to extract mathematical assets, establish the five-layer axiomatization structure, and propose a testability program.

Table 1: Complete Review of the Five-Paper Series and the Continuity Relationship of This Paper

Paper	Core Contribution	Position in This Paper
Tang (2026n)	Discovery of charm and bottom regions, activation operators, precision locking	Empirical basis for piecewise equations
Tang (2026o)	Strange region, precursor cascade, Theorems 1–3	Data source for CP violation origin
Tang (2026p)	Theorem 4, Theorem 5, Proposition 6	Theoretical basis for demotion of gauge field theory
This paper Tang (2026q)	Demotion, reconstruction, mathematical assets, five-layer structure, testability program	Concluding paper of the series

2. The Demotion of Gauge Field Theory

2.1 From Regime Boundaries to Gauge Groups: Established Conclusions

Tang (2026p) completed the proof of two theorems and one proposition. The Gauge Group Structure Theorem (Theorem 4): given that color charge is a meta-regime factor with a three-state structure, $SU(3)$ is the unique gauge group satisfying the conditions. The Direct Product Theorem (Theorem 5): the orthogonal independence of color charge and flavor quantum numbers necessarily leads to a direct product of gauge groups.

The proofs of these two theorems require no physical assumptions — the premises (the existence and orthogonal independence of the meta-regime factor) were confirmed in the more than 40 tests of the previous two papers, and the derivation logic is entirely based on the Cartan-Killing classification of Lie groups. Therefore, $SU(3) \times SU(2) \times U(1)$ is not assumed — it is the unique mathematical structure that inevitably emerges from the statistical structure of regime boundaries.

2.2 Formalized Statement of the Demotion

The demotion of gauge field theory can be stated as the following formalized proposition.

Proposition (The Demotion of Gauge Field Theory): Gauge field theory is not the most fundamental framework. It is the expression of regime boundaries in the language of group theory. The Factor Hierarchy Law is a deeper organizing principle than gauge field theory.

Argument:

1. Gauge field theory cannot derive the Factor Hierarchy Law. The core of gauge field theory is continuous symmetry — $SU(3)$ color symmetry, $SU(2)$ weak isospin symmetry, and $U(1)$ hypercharge symmetry.

The core of the Factor Hierarchy Law is discrete regime switching — the structural rupture of equations at the critical values of regime factors. Continuous symmetry cannot derive a piecewise function structure — the Lagrangian of gauge field theory is smooth along the mass axis and contains no mechanism for "regime switching." This assertion rests on the mathematical structure of gauge field theory itself, not on empirical facts.

A potential objection must be addressed: gauge field theory together with the Higgs mechanism (spontaneous symmetry breaking) does produce mass thresholds — does this count as "deriving regime switching"? It does not. The mass thresholds produced by the Higgs mechanism are parameter values in gauge field theory, not structural ruptures in the equations. Gauge field theory can describe interactions at each mass value, but it cannot explain why, at a specific mass, the equation coefficients undergo a 19-fold or –150-fold jump. If the Higgs mechanism could derive regime switching, it should be able to predict the 19-fold jump in the Mass coefficient at the charm quark threshold — but gauge field theory cannot produce this specific number. The Factor Hierarchy Law, through the Chow test, not only detects this jump but also precisely quantifies its magnitude. This specific number — 19-fold — is the statistical fingerprint of regime switching, which gauge field theory cannot derive from first principles.

2. The Factor Hierarchy Law can derive gauge field theory. The three-state structure of the meta-regime factor necessarily generates $SU(3)$ (Theorem 4), and orthogonal independence necessarily generates a direct product structure (Theorem 5).
3. Therefore, the Factor Hierarchy Law is a deeper organizing principle than gauge field theory.

It must be candidly stated that the assertion "gauge field theory cannot derive the Factor Hierarchy Law" depends on the mathematical fact that "continuous symmetry cannot generate a piecewise function." The Lagrangian of gauge field theory is smooth along the mass axis and contains no mechanism for "regime switching" — this is determined by the mathematical structure of gauge field theory, not by empirical facts. Hence, this assertion is mathematically rigorous, not an empirical induction.

2.3 Isomorphism with the Demotion of Universality in Tang (2026m)

The demotion of gauge field theory is structurally completely isomorphic with the "demotion of universality" in Tang (2026m). Tang (2026m) argued: universality is a mathematical convergence product in the thermodynamic limit ($L \rightarrow \infty$), while regime boundaries are the fundamental quantities in finite systems. This paper argues: gauge field theory is a mathematical structure in the continuous field theory limit, while regime boundaries are the fundamental quantities at finite energy scales.

Both share the same core logic: the "fundamental concepts" in traditional theories are mathematical products of idealized limits, while the Factor Hierarchy Law describes the real structures in finite systems. This isomorphism constitutes yet another piece of evidence for the Factor Hierarchy Law as a cross-disciplinary universal regularity.

2.4 Logical Connection Between the Demotion Argument and the Subsequent Theoretical Reconstruction

The above demotion argument establishes the meta-theoretical status of the Factor Hierarchy Law — it is not a "replacement" for the Standard Model but a "deriver" of the Standard Model. Gauge field theory is demoted to the expression of regime boundaries in the language of group theory.

The following three chapters, Chapter 3 through Chapter 5, will proceed from this meta-theoretical status to carry out a theoretical reconstruction of the Standard Model — reinterpreting three core concepts previously regarded as "fundamental": the statistical origin of CP violation, the regime function of the Higgs mechanism, and the statistical correspondence of virtual intermediate states. These three concepts are respectively regarded in the Standard Model as an "unsolved mystery," a "mass-origin mechanism," and a "mathematical tool" — within the framework of the Factor Hierarchy Law, they will be endowed with entirely new theoretical interpretations. These interpretations are not denials of the Standard Model but revelations of a deeper logic within the Standard Model: they are regarded as "fundamental" only because the perspective of gauge field theory cannot see regime boundaries as a deeper organizing principle.

3. The Statistical Origin of CP Violation

This chapter is the first part of the theoretical reconstruction of the Standard Model: the statistical origin of CP violation.

3.1 Review of Precursor Cascade Asymmetry

The precursor cascade is the transmission mechanism of regime switching along the mass axis. Tang (2026o) discovered and verified with precision quantification that the fluctuation transformation point (T_2) of each regime center is precisely the precursor boundary of the next center. Strange T_2 (2.1052 GeV) $\approx$ Charm T_1 (2.1000 GeV), deviation 0.0052 GeV. Charm T_2 (2.2865 GeV) = Bottom precursor (2.2865 GeV), deviation 0.0000 GeV.

Table 2: Interaction Effect Test of Cascade Asymmetry as a Regime Factor

Interaction Term	p-value	ΔR^2	Significant?
Mass $\times$ Cascade_Asymmetry	6.34×10^{-7}	0.0855	✓ Highly significant
Spin $\times$ Cascade_Asymmetry	1.27×10^{-2}	0.0494	✓ Significant
Isospin $\times$ Cascade_Asymmetry	1.59×10^{-2}	0.0486	✓ Significant

Note: Baseline model $R^2 = 0.4121$. All three interaction terms are significant. Cascade asymmetry modulates the exposure coefficients of all three executive factors on decay width — satisfying the defining characteristic of a regime factor.

Table 3: Detailed Results of the Cascade Asymmetry Chow Test

Cascade Pair	Chow F	p-value	Sample Size (Low/High)	Significant?
Strange→Charm	12.63	3.57×10^{-9}	101/105	✓ Highly significant
Charm→Bottom	0.88	4.76×10^{-1}	105/19	✗ Not significant

Note: Cascade asymmetry produces a significant Chow rupture in the Strange→Charm transmission but not in the Charm→Bottom transmission. This corresponds exactly to the magnitude of the precursor deviations — Strange→Charm deviation 0.0052 GeV (asymmetry significant), Charm→Bottom deviation 0.0000 GeV (perfectly symmetric).

3.2 The Statistical Definition of CP Violation

Within the framework of the Factor Hierarchy Law, CP violation can be given an entirely new statistical definition: **CP violation is the cumulative manifestation of precursor cascade asymmetry at**

higher mass scales. Within the Chow test framework, a non-zero deviation means that the positions of structural ruptures do not completely coincide — two regime boundaries that should be strictly symmetric exhibit an offset. This offset is statistically equivalent to "symmetry breaking."

Table 4: Current Data Comparison of Precursor Cascade Strength and CKM Matrix Elements

Cascade Pair	Cascade Deviation (GeV)	Cascade Strength	CKM Matrix Element	Coupling Strength
Strange→Charm	0.0052	192	$ V_{cs} \approx 0.975$	Strong coupling
Charm→Bottom	0.0000	>99999	$ V_{cb} \approx 0.041$	Weak coupling

Note: The correspondence between the deviations of the two cascade pairs and the CKM matrix elements is qualitatively consistent — the weaker the coupling, the more symmetric the cascade. However, the universality of this relationship requires verification through measurement of the bottom→top cascade deviation. With only two data points at present, it is insufficient to establish a quantitative regularity.

The statistical origin of CP violation lies in precursor cascade asymmetry. The complete verification of this proposition requires measurement of the bottom→top cascade deviation. Among the current two data points, the Strange→Charm deviation is non-zero (0.0052 GeV), and the Charm→Bottom deviation is zero (0.0000 GeV). If the bottom→top cascade deviation is also non-zero, then CP violation is the result of cascade asymmetry being reactivated and accumulated at specific cascade pairs. If the bottom→top cascade deviation is zero, then the origin of CP violation requires reconsideration.

It must be clarified that the deviations of the current two cascade pairs do not represent simple "linear accumulation" — the Strange→Charm deviation is non-zero (0.0052 GeV), while the Charm→Bottom deviation is zero (0.0000 GeV). This means that cascade asymmetry is not progressively accumulated from the first cascade onward but may be "reactivated" at specific cascade pairs. The physical intuition is: the Strange→Charm cascade involves transmission between the strange quark and the charm quark, while the Charm→Bottom cascade involves transmission between the charm quark and the bottom quark. The physical mechanisms of the two are different — strange and charm both belong to the "light" heavy quarks, while charm and bottom straddle the switching-type boundary from "rule-resetting type" to "direction-reversal type" (see Tang, 2026o, Theorem 2). Cascade asymmetry may be reactivated at cascade pairs that cross switching-type boundaries.

The logical relationship between "reactivation" and "accumulation" must be clarified: after cascade asymmetry is reactivated at a switching-type boundary, it may further accumulate in subsequent cascades. If the bottom→top cascade deviation is larger than the Strange→Charm deviation (0.0052 GeV), this supports the accumulation hypothesis — asymmetry amplifies with increasing mass scale. If it falls between 0.0000 GeV and 0.0052 GeV, this supports the reactivation hypothesis — asymmetry is partially restored at the switching-type boundary. The two hypotheses can be distinguished through measurement of the bottom→top cascade deviation.

3.3 Relationship with the CP-Violating Phase of the CKM Matrix

CP violation in the CKM matrix is described by a single phase $\delta_{13} \approx 1.2$ rad (Kobayashi & Maskawa, 1973). If precursor cascade asymmetry is the statistical origin of CP violation, then a quantitative relationship should exist between cascade asymmetry and δ_{13} . The current two data points are insufficient to establish this relationship, but a testable prediction can be made: the measured value of the bottom→top cascade deviation will exhibit a quantitative correspondence with the value of δ_{13} .

4. The Regime Function of the Higgs Mechanism

This chapter is the second part of the theoretical reconstruction of the Standard Model: the regime function of the Higgs mechanism.

4.1 From Mass Origin to Regime Boundary Generation

The empirical discovery of Tang (2026n) provides an entirely new perspective on the Higgs mechanism. Color charge was confirmed as the meta-regime factor. In the Standard Model, color charge originates from the Yukawa coupling between quarks and the Higgs field — the Higgs mechanism is the physical source of color charge.

If color charge is the meta-regime factor, and the Higgs mechanism is the source of color charge, then the essential function of the Higgs mechanism is "to generate regime boundaries." This inference can be articulated at two levels.

Level 1 (Inference supported by evidence): The Higgs mechanism generates regime boundaries. The logical chain is: Yukawa coupling between Higgs field and quarks → quarks acquire mass → color charge emerges as the meta-regime factor → regime boundaries are generated → the mass spectrum is partitioned into four discrete regime centers.

Level 2 (Philosophical interpretation): Conferring mass is a by-product of regime boundary generation. In this logical chain, the origin of mass is not the primary function of the Higgs mechanism — the primary function is to generate regime boundaries. This interpretation is not strongly asserted; it is positioned as "a logical inference within the framework of the Factor Hierarchy Law."

4.2 Supplementary Derivations

4.2.1 Asymptotic Freedom

The strong interaction becomes weaker at short distances (high energy scales). In regime boundary theory, the density of regime boundaries decreases at high energy scales — the four regime centers are densely distributed in the 1.5–10 GeV range, while the bottom→top spacing (162.4 GeV) far exceeds the charm→bottom spacing (7.17 GeV).

This trend can be quantified: strange→charm spacing / strange mass ≈ 0.53 , charm→bottom spacing / charm mass ≈ 3.13 , bottom→top spacing / bottom mass ≈ 17.2 . This ratio sequence (0.53, 3.13, 17.2) clearly demonstrates the trend of regime boundary density decreasing sharply with increasing energy scale. The equation structure becomes increasingly stable — this manifests as the logarithmic decay of the coupling constant.

This interpretation is qualitative; its quantitative verification path is: compare Chow F density at different energy scales; at higher energy scales (such as LHC data), fewer structural breaks should be detected. If Chow F density exhibits a logarithmic decay trend with increasing energy scale, a testable quantitative correspondence exists with the coupling constant decay rate predicted by the β -function.

4.2.2 Color Confinement

Quarks cannot exist in isolation. Color charge is the meta-regime factor — it divides particles into the color-charged regime and the color-neutral regime, which are logically mutually exclusive. Color-charged particles (quarks) can never exist as free particles in the color-neutral regime. Color confinement is not a dynamical result (not "the strong interaction is too strong for them to escape") but a logical result ("color-charged particles cannot exist in the color-neutral regime").

This interpretation predicts that any process attempting to transfer quarks from the color-charged regime to the color-neutral regime will produce detectable structural breaks at the regime boundary. Data from quark-gluon plasma in high-energy heavy-ion collisions may provide an experimental platform for testing this prediction.

4.2.3 Spontaneous Chiral Symmetry Breaking

The mass of a nucleon is much larger than the sum of the masses of its constituent quarks. In the charm-region piecewise decay equation, the Mass coefficient differs by 19-fold across the regime boundary. This means that the exposure coefficient of the executive factor "Mass" can be dramatically altered when crossing regimes. When quarks enter the strongly bound state (constituent mass) from the weakly bound state (current mass), their effective mass is redefined by the regime boundary — this is the statistical root of the enormous difference between "current mass" and "constituent mass."

This interpretation predicts that, at the cross-regime boundary, there exists a proportionality factor positively correlated with the Mass coefficient jump between the constituent mass of hadrons and the current mass of quarks. The quark condensate value computed by lattice QCD may provide independent verification.

5. The Statistical Correspondence of Virtual Intermediate States

This chapter is the third part of the theoretical reconstruction of the Standard Model: the statistical correspondence of virtual intermediate states.

5.1 The Functional Isomorphism Between the Precursor Cascade and Virtual States

Virtual particles in quantum field theory — particles that appear as internal lines in Feynman diagrams and do not satisfy the on-shell condition — are core concepts of perturbative calculations. The precursor cascade provides a data-driven statistical counterpart for this concept. After the bound states of the charm

quark complete their fluctuation transformation at 2.29 GeV (T_2), they provide a precursor signal for the regime switch of the bottom quark at 9.46 GeV.

Functional isomorphism:

- Virtual charm quark pairs: appear as internal lines in Feynman diagrams, transmit quantum fluctuations, do not satisfy on-shell conditions.
- Precursor cascade: appears as the coincidence of T_2 and the precursor on the mass axis, transmits regime-switching signals, manifests as structural breaks in the Chow test.

5.2 Statistical Interpretation and Testable Prediction

The precursor cascade does not depend on perturbative expansions or the presuppositions of Feynman diagrams — it emerges entirely automatically from the structural breaks in the data. If a deep correspondence exists between the precursor cascade and virtual intermediate states, then a positive correlation should exist between the strength of the precursor cascade and the contribution of virtual charm quark pairs in bottom quark decays.

The physical intuition for predicting a positive correlation rather than an anti-correlation is: the more symmetric the cascade (the smaller the deviation), the higher the efficiency of precursor signal transmission — precise alignment of two regime boundaries means that regime-switching information can be transmitted from one center to the next without attenuation. In quantum field theory, the essential function of virtual intermediate states is precisely the efficient transmission of quantum fluctuations — virtual charm quark pairs serve as information transmission media in bottom quark decays. Therefore, the higher the cascade precision (the larger $1/\text{deviation}$), the larger the corresponding virtual state contribution should be. If experimental observations show an anti-correlation (higher precision, smaller virtual state contribution), the correspondence between the precursor cascade and virtual intermediate states would require reconsideration.

It must be candidly stated that the strength of the precursor cascade ($1/\text{deviation}$) measures "the precision of alignment between two regime boundaries," while the production rate of virtual charm quark pairs measures "the frequency with which the intermediate state is used." Although these two quantities should logically be positively correlated, they have different dimensions and different physical meanings, and their precise functional relationship (linear or nonlinear) needs to be determined by experimental data.

A more specific quantitative correspondence can be formulated as follows: the ratio of the precursor cascade deviation (0.0052 GeV) to the charm quark mass (approximately 1.27 GeV) is approximately 0.4%. In bottom quark decays, the contribution of virtual charm quark pairs can also be measured by a similar dimensionless ratio — the ratio of the virtual charm quark pair production rate to the total decay rate. If a

quantitative correspondence exists between these two ratios, it would constitute decisive verification of the "statistical correspondence," transcending the current "functional isomorphism" argument.

This prediction can be tested using experimental data from LHCb or Belle II.

5.3 Epistemological Significance of the Statistical Interpretation

This is structurally completely isomorphic with the argument in Tang (2026m) that "universality is a mathematical convergence product in the thermodynamic limit, while regime boundaries are the fundamental quantities in finite systems." Virtual states are mathematical tools in perturbative expansions; the precursor cascade is a fundamental structure in finite systems.

6. Mathematical Assets: Quantitative Formulas Emerging from Data

Based on all empirical findings from the previous three papers, this section extracts directly usable mathematical assets.

Table 5: Summary of Core Mathematical Assets of the MRSD Algorithm

Asset No.	Name	Formula	Empirical Basis	Functional Equivalence
MA1	Charm-region piecewise decay equation	See §6.1	Tang (2026n) Table 9	Equivalent to 3D Ising piecewise control function
MA2	Bottom-region piecewise decay equation ($W \times M$ activated)	See §6.2	Tang (2026n) Table 9	Same as above
MA3	Precursor cascade formalization	$T_2(C_i) = T_{\text{precursor}}(C_i + 1)$	Tang (2026o) Table 5	Mathematical formulation of cascade transmission
MA4	Universal transition zone width formula	$W_{\text{cross}} = \Sigma W_{\text{internal}} + \Delta$	Tang (2026o) Theorem 1	Topological constraint on regime switching in finite systems
MA5	Switching type criterion	$r = \Delta\alpha/\Delta\beta$; $r < 2$ or $r > 5$	Tang (2026o) Theorem 2	Mathematical basis for MRSD three-category classification
MA6	Five-layer derivation	Axioms $\rightarrow$ Norms $\rightarrow$	This paper	Complete derivation

Asset No.	Name	Formula	Empirical Basis	Functional Equivalence
	chain	Algorithm → Theorems → Theory	§7	chain of the theoretical system

6.1 Charm-Region Piecewise Decay Equation

$$Width(M, S, I) = \begin{cases} -0.1169 + 0.1558 \cdot M - 0.0155 \cdot S + 0.0841 \cdot I, & M < 2.09 GeV [Regime I: Pre - Charm] \\ \mathcal{F}_{transition}(M, S, I), & 2.09 \leq M \leq 2.29 GeV [Regime II: Charm Transition Zone] \\ -0.0477 + 0.0082 \cdot M + 0.0466 \cdot S + 0.0232 \cdot I, & M > 2.29 GeV [Regime III: Post - Charm] \end{cases}$$

The Mass coefficients of Regime I and Regime III differ by a factor of 19 (0.1558 vs. 0.0082). In the Standard Model, a significant change in decay dynamics indeed occurs at the charm quark threshold — particles in the pre-charm region decay primarily through the strong interaction, while particles in the post-charm region can decay through the weak interaction, with the dependence of decay width on mass significantly weakening.

6.2 Bottom-Region Piecewise Decay Equation (W×M Activated Form)

$$W \times M(M, S, I) = \begin{cases} 0.10 + 0.02 \cdot M + 0.06 \cdot S + 0.11 \cdot I, & M < 6.65 GeV [Regime I: Pre - Bottom] \\ \mathcal{F}_{transition}(M, S, I), & 6.65 \leq M \leq 10.16 GeV [Regime II: Bottom Transition Zone] \\ -15.33 + 1.51 \cdot M - 0.10 \cdot S - 0.50 \cdot I, & M > 10.16 GeV [Regime I: Pre - Bottom] \end{cases}$$

The intercepts of Regime I and Regime III differ by a factor of −150. The high-mass segment of the bottom region consists mainly of the Y family (bottomonium), whose decay widths are much larger than those of bottom mesons and bottom baryons. The sample size for Regime III is only 12 (mainly the Y family), and the statistical uncertainty of the coefficient estimates is relatively large — but the direction and magnitude of the intercept jump are unequivocal.

6.3 Formalized Expression of the Precursor Cascade

$$T_2(C_i) = T_{precursor}(C_{i+1});, \quad i = 1, 2, 3$$

Current empirical status: i = 1 deviation 0.0052 GeV, i = 2 deviation 0.0000 GeV, i = 3 deviation 162.4 GeV (preliminary estimate due to insufficient sample size in the top region).

6.4 Universal Formula for Transition Zone Width

$$W_{\text{cross}}(C_i \rightarrow C_{i+1}) = W_{\text{internal}}(C_i) + \Delta_{i,i+1} + W_{\text{internal}}(C_{i+1})$$

When $\Delta = 0$, $W_{\text{cross}} = W_{\text{internal}}(C_i) + W_{\text{internal}}(C_{i+1})$. Overlapping transition zones are governed by the minimum rule: $W_{\text{overlap}} = \min(W_i, W_{i+1})$.

6.5 Intercept/Slope Ratio Criterion for Switching Types

$$r = \frac{\Delta\alpha}{\Delta\beta} \begin{cases} < 2 & \text{Rule-Resetting Type} \\ > 5 & \text{Direction-Reversal Type} \end{cases}$$

$r = \frac{\Delta\alpha}{\Delta\beta} \begin{cases} < 2 & \text{Rule-Resetting Type} \\ > 5 & \text{Direction-Reversal Type} \end{cases}$

Table 6: Complete Summary of Intercept/Slope Ratios for Nine Data Points Across Six Regime Centers

Regime Center	Threshold	Intercept/Slope Ratio	Classification	Covariance Rupture (p)
Strange	ICSS 1.55	1.49	Rule-resetting	0.318
Strange	Bai-Perron $T_1=2.0200$	1.95	Rule-resetting	0.184
Strange	Bai-Perron $T_2=2.1052$	1.14	Rule-resetting	0.284
Strange	Chow 2.29	0.77	Rule-resetting	—
Charm	Chow 2.29	0.77	Rule-resetting	0.229
Bottom	Y(1S) 9.46	5.73	Direction-reversal	0.005
Bottom	$W \times M^2$ Y(2S) 10.02	5.53	Direction-reversal	—
Bottom	Hidden T_1 6.65	6.27	Direction-reversal	—
Bottom	Hidden T_2 10.16	5.48	Direction-reversal	—

The above six mathematical assets constitute the core output of the algorithm layer and the theorem-and-proposition layer in the five-layer axiomatization structure. The piecewise equations and

cascade formula are direct products of the MRSD algorithm — input data, execute structural optimization, output regime matrix. The transition zone width formula and switching type criterion are the formalized results of the theorem-and-proposition layer — rigorous mathematical derivations from axioms. The following §7 integrates these assets into a complete five-layer axiomatization structure, displaying the complete derivation chain from axioms to theorems to theory.

7. The Five-Layer Axiomatization Structure

7.1 Structural Overview

Based on all empirical findings, mathematical theorems, and theoretical derivations from the previous four papers, the Factor Hierarchy Law, as a complete scientific theoretical system, exhibits a rigorous five-layer axiomatization structure.

Table 7: Overview of the Five-Layer Axiomatization Structure of the Factor Hierarchy Law

Layer	Name	Core Content	Function
First Layer	Law Layer (Axiom System)	Three axioms	Describes the universal logic of regime switching in all complex systems
Second Layer	Norm Layer (Operational Standards)	Six functional dimensions	Transforms axioms into operable procedural standards
Third Layer	Algorithm Layer (Operational Implementation)	MRSD three-step logic	Transforms norms into executable mathematical procedures
Fourth Layer	Theorem and Proposition Layer (Mathematical Proof)	Five theorems + one proposition	Derives undeniable conclusions from axioms through rigorous deduction
Fifth Layer	Theory Layer (Physical Interpretation)	Eight inferences	Integrates theorems into a self-consistent physical picture

This five-layer structure did not come into being all at once — it emerged gradually through verification in four completely independent disciplines. The Factor Hierarchy Law originated in finance (Tang, 2026g) — the China-U.S. yield spread as a regime factor triggered a regime switch in global asset pricing equations near spread ≈ 0 . In astronomy, stellar spectral type as a regime factor triggered a regime

switch in transit depth equations at 4762 K — the Tang Break, with all five independent physical dimensions undergoing structural rupture (Tang, 2026h, 2026i, 2026j). The MRSD algorithm discovered hidden structures never revealed by traditional finite-size scaling analysis on the 3D Ising model — double-peak fission, transition regimes, and precursor boundaries (Tang, 2026l). The demotion of universality completed the rigorous proof of regime boundaries as the fundamental quantities in finite systems across five phase-transition models (Tang, 2026m). Finally, this methodological system, in the core data of the Standard Model of particle physics — from the discovery of the charm and bottom regions (Tang, 2026n), to the confirmation of the strange region and the precursor cascade (Tang, 2026o), to the inevitable emergence of the gauge group structure (Tang, 2026p), to the theoretical elevation and system legislation of this paper — has completed a complete closed loop from universal regularities to specific verification to theoretical elevation.

This demotion logic is structurally completely isomorphic with the demotion of universality in Tang (2026m), demonstrating that "demoting the mathematical products of idealized limits to statistical approximations" is the core argumentative mode of the cross-disciplinary unification of the Factor Hierarchy Law — it demotes universality in statistical physics to an approximation in the thermodynamic limit, and demotes gauge field theory in particle physics to an expression in the continuous field theory limit. Five theorems, eight inferences, five predictions — every one has been cross-validated by at least two independent tools, and every one has been derived through rigorous deduction from axioms.

7.2 The Fourth Layer: Theorem and Proposition Layer

Table 8: Summary of the Five Theorems and One Proposition

Theorem/Proposition	Name	Core Statement	Evidence Level	Source
Theorem 1	Transition Zone Width Theorem	Inter-center transition zone width equals sum of two internal widths plus gap	Theorem	Tang (2026o)
Theorem 2	Switching Type Dichotomy Theorem	$r < 2$ or $r > 5$; the intermediate zone is unstable	Theorem	Tang (2026o)
Theorem 3	Switching Type Intrinsic Property Theorem	All thresholds within the same regime center have the same type	Theorem	Tang (2026o)
Theorem 4	Gauge Group Structure Theorem	The three-state structure of a meta-regime factor necessarily generates SU(3)	Theorem	Tang (2026p)
Theorem 5	Direct Product Theorem	Orthogonally independent meta-regime factors necessarily lead to direct product of gauge groups	Theorem	Tang (2026p)
Proposition 6	Three-Generation Proposition	The number of heavy quarks is limited to four by the topological constraint of transition zone width	Proposition (to be elevated)	Tang (2026p)

Note: Theorems 1 through 5 are rigorous mathematical derivations from axioms. The argumentation of Proposition 6 depends on an empirical trend constituted by three data points and the independent experimental exclusion by the LHC; its elevation to a theorem requires confirmation of the internal double-peak fission of the top region and tests from experiments at higher energy scales.

7.3 The Fifth Layer: Theory Layer

Table 9: Summary of Eight Physical Inferences

Inference	Core Content	Derivation Basis	Source
Inference 1	The Standard Model is a hierarchical system carved by four regime centers	Theorems 1–5	Tang (2026n, 2026o)
Inference 2	Gauge field theory is the expression of regime boundaries in the language of group theory	Theorems 4, 5	Tang (2026p)
Inference 3	The Factor Hierarchy Law is a deeper organizing principle than gauge field theory	Theorems 4, 5 + causal direction argument	Tang (2026p) + this paper
Inference 4	CP violation is the cumulative manifestation of precursor cascade asymmetry	Axiom 4 + cascade data	This paper
Inference 5	The essential function of the Higgs mechanism is to generate regime boundaries	Axiom 1 + color charge confirmation	This paper
Inference 6	Virtual intermediate states are the quantum expression of precursor signals	Axiom 4 + cascade data	This paper
Inference 7	Activation operator selectivity is regime selection	11-operator exhaustive stratification	Tang (2026n)
Inference 8	The number of heavy quarks is determined by the topological constraint of transition zone width	Theorem 1 + Proposition 6	Tang (2026p)

7.4 New Forms of Factor Hierarchy Discovered in Particle Physics

Table 10: Summary of New Forms of the Factor Hierarchy Law Discovered in Particle Physics

New Form	Core Characteristics	Relationship to Original Classification of Tang (2026g)	Discovery Paper
Cascade Regime System	Multiple regime centers connected by a precursor cascade; transmission between layers occurs through boundary alignment rather than interaction effect modulation	Entirely new form not covered by original classification	Tang (2026o)
Activation Operator Dual-Identity Extension	The same factor plays different roles through different mathematical transformations in different regimes	Extension of "dual-identity factor" in original classification	Tang (2026n)
Processual Regime Factor	Modulates the transmission efficiency of regime switching, rather than "which set of rules applies"	Entirely new form not covered by original classification	Tang (2026o)
Orthogonal Multi-Regime-Factor Parallelism	Color charge and flavor quantum numbers are orthogonally independent and regulate in parallel	Deepening of "multi-regime-factor" in original classification	Tang (2026n, 2026o)

Note: The cascade regime system is the most important new form revealed by the Factor Hierarchy Law in particle physics — it is an entirely new mode of factor-hierarchy organization, in which transmission between layers occurs through boundary-aligned cascades rather than through the modulation of interaction effects. This is essentially different from the mechanism in the original classification whereby "regime factors modulate the exposure coefficients of executive factors through interaction effects."

8. Testability Program

Table 11: Summary of Five Testable Predictions and Their Verification Paths

Prediction	Core Content	Verification Path	Falsification Condition
Prediction 1	If a fourth generation of heavy quarks is discovered in the future, a structural break detectable by the Chow test must exist at its mass	Experiments at higher energy scales	Discovery of a fourth generation of heavy quarks without a corresponding regime-switching center
Prediction 2	The switching type of a fourth generation of heavy quarks will fall into the stable regions of $r < 2$ or $r > 5$	Determination of the intercept/slope ratio of the fourth-generation regime center	The switching type falls in the intermediate zone of 2–5
Prediction 3	A quantitative relationship exists between precursor cascade strength and the off-diagonal elements of the CKM matrix	Confirmation of internal double-peak fission of the top region + measurement of bottom→top cascade deviation	No correlation between the strengths of the three cascade pairs and the CKM matrix elements
Prediction 4	The neutrino mass hierarchy is a regime boundary, not a continuous parameter	Chow test on neutrino oscillation data	No structural break on the neutrino mass axis
Prediction 5	The regularities of transition zone width similarly hold in other complex systems	Cross-disciplinary verification	Not holding in any other system

9. Conclusion

This paper is the concluding piece of the regime audit series on the Standard Model of particle physics, completing four core missions.

First, it completes the demotion of gauge field theory. It demonstrates that gauge field theory is not the most fundamental framework — it is the expression of regime boundaries in the language of group theory. The Factor Hierarchy Law is a deeper organizing principle than gauge field theory. The demotion argument is structurally completely isomorphic with the demotion of universality in Tang (2026m).

Second, it completes the theoretical reconstruction of the Standard Model. The statistical origin of CP violation lies in precursor cascade asymmetry — cascade asymmetry may be reactivated at cascade pairs that cross switching-type boundaries, with its complete verification requiring measurement of the bottom→top cascade deviation. It is predicted that the bottom→top cascade deviation will fall between 0.0000 GeV and 0.0052 GeV. The essential function of the Higgs mechanism is to generate regime boundaries — Level 1 is an inference supported by evidence, and Level 2 is a philosophical interpretation. Virtual intermediate states are the quantum expression of precursor signals — a positive correlation is predicted between precursor cascade strength and the contribution of virtual charm quark pairs.

Third, it extracts directly usable mathematical assets. The charm-region piecewise decay equation (Mass coefficient differing by a factor of 19), the bottom-region piecewise decay equation (intercept differing by a factor of −150), the formalized expression of the precursor cascade, the universal formula for transition zone widths, the intercept/slope ratio criterion for switching types, and the five-layer derivation chain.

Fourth, it formalizes the five-layer axiomatization structure and proposes a testability program. Law layer → Norm layer → Algorithm layer → Theorem and Proposition layer → Theory layer. Five testable predictions await independent verification by future experiments. The four new forms discovered in particle physics enrich the original classification framework of the Factor Hierarchy Law.

The audit of the Factor Hierarchy Law in the Standard Model of particle physics, after more than 40 tests, six structural break tools, 11 activation operators, a 2569-fold activation effect, 10^{-4} GeV precision deviations, five mathematical theorems, one proposition, eight physical inferences, and five testable predictions, is hereby completed.

Data Availability Statement

This paper is a theoretical reinterpretation and system integration paper. All empirical data come from Tang (2026n) and Tang (2026o). The complete proofs of all mathematical theorems can be found in Tang (2026o) and Tang (2026p). The data are completely open, and any researcher can independently reproduce all results.

References

[1] Bai, J., & Perron, P. (1998). Estimating and testing linear models with multiple structural changes. *Econometrica*, 66(1), 47–78. DOI: 10.2307/2998540

- [2] Chow, G. C. (1960). Tests of equality between sets of coefficients in two linear regressions. *Econometrica*, 28(3), 591–605. DOI: 10.2307/1910133
- [3] Kobayashi, M., & Maskawa, T. (1973). CP-violation in the renormalizable theory of weak interaction. *Progress of Theoretical Physics*, 49(2), 652–657. DOI: 10.1143/PTP.49.652
- [4] Mayo, D. G. (1996). *Error and the Growth of Experimental Knowledge*. University of Chicago Press. DOI: 10.7208/chicago/9780226511993.001.0001
- [5] Onsager, L. (1944). Crystal statistics. I. A two-dimensional model with an order-disorder transition. *Physical Review*, 65(3–4), 117–149. DOI: 10.1103/PhysRev.65.117
- [6] Particle Data Group. (2025). Review of Particle Physics. *Progress of Theoretical and Experimental Physics*, 2025(8), 083C01. DOI: 10.1093/ptep/ptae106
- [7] Popper, K. (1959). *The Logic of Scientific Discovery*. Hutchinson. DOI: 10.1063/1.3060577
- [8] Tang, S. (2026g). Testability: Factor Hierarchy Theory and Scientific Methodology for Multi-Factor Complex Systems—Complete Empirical Evidence from Global Asset Pricing and a Cross-Disciplinary Application Paradigm. Zenodo, DOI: 10.5281/zenodo.21104307.
- [9] Tang, S. (2026h). Planetary Radius as a Regime-Dependent Relational Quantity: Evidence for a Paradigm Defect in Exoplanet Radius Measurement—A Cross-Disciplinary Empirical Test in Astronomy Based on Factor Hierarchy Theory and the Four-Test Method. Zenodo, DOI: 10.5281/zenodo.21105475.
- [10] Tang, S. (2026i). Self-Consistency Rupture in Stellar Atmosphere Models: The Second Astronomical Empirical Case for the Factor Hierarchy Law—Evidence from Internal Cross-Validation Based on the Four-Test Method. Zenodo, DOI: 10.5281/zenodo.21127665.
- [11] Tang, S. (2026j). The Stellar Main-Sequence Regime Boundary: Five-Dimensional Independent Verification of the 4762 K Critical Point and the Systematic Establishment of the Factor Hierarchy Law—Cross-Dimensional Integrated Evidence Based on the Four-Test Method. Zenodo, DOI: 10.5281/zenodo.21128329.
- [12] Tang, S. (2026k). Information-Theoretic Duality Between Regression Topology and Statistical Mechanics—A Data-Driven Equivalence of the Ehrenfest Classification Validated on the 2D Ising Model as the Physics Cornerstone for the Factor Hierarchy Law. Zenodo, DOI: 10.5281/zenodo.21154577.
- [13] Tang, S. (2026l). Double-Peak Fission: The Hidden Structure of Finite-Lattice Phase Transitions in the 3D Ising Model—Verification of the Factor Hierarchy Law on an NP-Complete System: Double-Peak

Fission, Transition Regime, Piecewise Control Function, and Multi-Regime State Decoupling Algorithm. Zenodo, DOI: 10.5281/zenodo.21257777.

[14] Tang, S. (2026m). The Reduction of Universality: A Regime Boundary Theory for Finite-System Phase Transitions—Cross-Model Evidence for Bimodal Splitting, the Precursor Regime Boundary, and the Tool-Dependence Theorem. Zenodo, DOI: 10.5281/zenodo.21258474.

[15] Tang, S. (2026n). Regime Audit of the Standard Model of Particle Physics by the Factor Hierarchy Law— Activation Operators, Dual-Center Regimes, and Precision Locking of Charm/Bottom Quark Thresholds(I) Zenodo, DOI: 10.5281/zenodo.21276887.

[16] Tang, S. (2026o). Strange Region Discovery, Precursor Cascade, and the Transition Zone Width Theorem— Four-Tool Cross-Validation, Cascade Asymmetry Confirmation, and Mathematical Proof of the Switching Type Dichotomy(II) Zenodo, DOI: 10.5281/zenodo.21277300.

[17] Tang, S. (2026p). Gauge Group Structure Theorem, Direct Product Theorem, and the Three-Generation Proposition— The Inevitable Emergence of $SU(3) \times SU(2) \times U(1)$ from Regime Boundaries and the Topological Constraint on the Number of Heavy Quarks(III) Zenodo, DOI: 10.5281/zenodo.21291670.

[18] Wilson, K. G. (1971). Renormalization group and critical phenomena. *Physical Review B*, 4(9), 3174–3205. DOI: 10.1103/PhysRevB.4.3174

[19] Yang, C. N., & Mills, R. L. (1954). Conservation of isotopic spin and isotopic gauge invariance. *Physical Review*, 96(1), 191–195. DOI: 10.1103/PhysRev.96.191